Autonomous Maritime Forensic Intelligence: Strategic Gap Analysis and Algorithmic Architecture for Satellite-Aided Hydrodynamic Oil Pollution Attribution

Operational Context and the Marine Surveillance Deficit

The expansion of international maritime trade has intensified the incidence of operational marine hydrocarbon discharges, encompassing deliberate oily bilge pumping, tank-washing residues, and bunker fuel purging across critical sea lanes[1]. In high-density shipping corridors, including the International Shipping Lanes traversing the northern Indian Ocean and the Exclusive Economic Zone of India, illicit operational discharges cause progressive environmental degradation to marine ecosystems and coastal assets[3]. Historically, coastal states and maritime law enforcement authorities have relied on aerial surveillance patrols, incidental vessel reports, and coastal radar installations[1]. These conventional methods face spatial limitations, high flight costs, operational latency, and intermittent revisit intervals, creating surveillance gaps across deep-sea environments[1].

The availability of open-access Earth Observation satellite constellations—particularly the European Space Agency Sentinel-1 Synthetic Aperture Radar (SAR) mission—alongside terrestrial and satellite-relayed Automatic Identification System (AIS) vessel tracking networks, provides the observational foundation for scalable maritime surveillance[1]. However, translating raw microwave backscatter damping and asynchronous vessel transponder broadcasts into scientifically defensible, actionable investigative attribution remains a significant technological challenge[1]. Problem Statement SIH26143 articulates the need for an automated investigative platform capable of detecting oil slicks on the ocean surface, reconstructing historical transport dynamics, filtering extraneous traffic, and pinpointing the responsible vessel for maritime interdiction[1].

A standard baseline pipeline that couples a basic convolutional neural network detector to a deterministic Lagrangian particle model and identifies the nearest AIS-transmitting vessel is inadequate for operational deployment[1]. Ocean surface dynamics exhibit turbulent dispersion and spatial divergence, numerical ocean circulation models carry boundary layer and interpolation errors, and transponder records are frequently degraded by signal attenuation, co-channel packet collisions, or intentional shutoffs[1]. Furthermore, radar backscatter damping is non-unique, producing look-alike signatures from natural biogenic slicks, low-wind zones, and

atmospheric phenomena[2]. Consequently, an operational platform must transition from an exploratory monitoring dashboard into an evidentiary decision-support system that models physical uncertainty, evaluates contradictory evidence, and provides tamper-evident data integrity[1].

Comparative Assessment of Global and Regional Surveillance Baselines

Establishing a technically grounded architecture requires a structured assessment of the primary platforms currently deployed internationally and regionally:

- **SkyTruth Cerulean:** An open-source system utilizing Sentinel-1 SAR imagery and deep convolutional neural networks (predominantly U-Net and ResNet backbones) to segment marine slicks and correlate them with global AIS vessel trajectories and offshore infrastructure[1]. Cerulean establishes vessel associations by computing spatial proximity, temporal overlap, and geometric alignment between vessel tracks and slick polygons[1]. While highly effective for macroscopic domain awareness, uncertainty-bounded attribution stability metrics, physical bidirectional forward validation, and adversarial counter-evidence scoring are not identified as publicly documented core capabilities in the sources reviewed[1].
- **EMSA CleanSeaNet:** An operational maritime pollution monitoring and alerting service operated by the European Maritime Safety Agency across European waters[1]. CleanSeaNet ingests multi-sensor Earth Observation data, detects slicks, tracks vessel traffic, and incorporates meteorological and oceanographic models to forecast spill movement and execute backward drift simulations[1]. While CleanSeaNet represents a mature operational infrastructure, it functions as a restricted alerting framework for member states rather than an open, reproducible forensic intelligence system that exposes transparent mathematical attribution proofs[1].
- **INCOIS OOSA:** The Online Oil Spill Advisory system operated by the Indian National Centre for Ocean Information Services[1]. OOSA couples atmospheric boundary forcing with numerical ocean hydrodynamic circulation models, such as regional NEMO configurations, utilizing trajectory engines like GNOME to project spill trajectories up to 96 hours in advance for the Indian Coast Guard[1]. OOSA is engineered for forward disaster response and boom deployment optimization; automated retrospective vessel attribution, dark vessel extraction, and multi-hypothesis AIS trajectory reconciliation are not identified as publicly documented core capabilities in the sources reviewed[1].
- **Global Fishing Watch:** Provides vessel intelligence by analyzing global AIS data, identifying coverage gaps, detecting dark vessels through radar-to-AIS cross-referencing, and matching vessel identities across public registries[1]. Global Fishing Watch is dedicated to fisheries management and marine protected area enforcement; native SAR oil slick segmentation models, ocean hydrodynamic drift modules, and petroleum weathering simulation engines are not identified as publicly documented core capabilities in the sources reviewed[1].

Capability Dimension	SkyTruth Cerulean	EMSA CleanSeaNet	INCOIS OOSA	Global Fishing Watch	Arpinia Odyssey Framework
SAR Slick Detection	Deep Learning (U-Net/ResNet)[1]	Operational Multi-SAR[1]	Not identified as a core capability[1]	Not identified as a core capability[1]	Multi-Architecture Ensemble[9]
Look-Alike Filtering	Heuristic & Contextual[1]	Expert Operator Review[1]	Not identified as a core capability[1]	Not identified as a core capability[1]	MetOcean & Textural Confidence Engine[2]
Boundary Uncertainty	Discrete Binary Mask[1]	Hard Edge Contours[1]	Not identified as a core capability[1]	Not identified as a core capability[1]	Epistemic Dropout & Threshold Sweep[1]
Drift Modeling	Spatial Buffers[1]	Lagrangian / Eulerian Drift[1]	Hydrodynamic Forecasting[1]	Not identified as a core capability[1]	Stochastic Lagrangian OpenDrift[7]
Attribution Logic	Geometric Association[1]	Deterministic Hindcast[1]	Not identified as a core capability[1]	Vessel Identity Matching[1]	Multi-Factor Evidentiary Score[1]
Stability Quantification	Not identified as a core capability[1]	Not identified as a core capability[1]	Not identified as a core capability[1]	Not identified as a core capability[1]	Monte-Carlo Convergence Stability[1]
Adversarial Evidence	Not identified as a core capability[1]	Operator Assessed[1]	Not identified as a core capability[1]	Not identified as a core capability[1]	Supporting vs. Contradictory Alibis[1]
Validation Mechanism	One-Way Match[1]	Operational Feedback[1]	Drifter Calibration[1]	Multi-Sensor Match[1]	Bidirectional Drift

			[3]		Validation[1]
Dark Vessel Analysis	AIS Gaps Flagged[1]	Radar Target Extraction[1]	Not identified as a core capability[1]	Advanced Gap Analytics[1]	Kinematic Reachability & SAR Fusion[1]
Forensic Deliverable	Web Layer & GeoJSON[1]	Restricted Alert Dossier[1]	Disaster Advisory Bulletin[1]	Data API Export[1]	Tamper-Evident Evidence JSON[1]

Comprehensive Typology of Research Gaps in Source Attribution

A systematic investigation of the literature reveals core structural and physical challenges in maritime pollution surveillance, establishing a distinction between deficiencies targeted for resolution and open research frontiers.

Gaps Targeted for Resolution by the Proposed Architecture

- **Mitigation of Look-Alike False Positives:** Addressing radar backscatter ambiguity by integrating multi-channel MetOcean wind context, GLCM textural metrics, and boundary gradient analysis without naive binary cuts[2].
- **Spill Boundary Uncertainty Quantification:** Eliminating rigid binary mask assumptions by modeling boundary gradients and epistemic model variance, passing continuous spatial confidence envelopes to downstream particle engines[1].
- **Cross-Scene Model Generalization:** Overcoming single-scene overfitting by validating segmentation models across diverse geographic basins, sea states, and radar incidence angles[8].
- **Resolution of Deterministic Hindcasting Fragility:** Eliminating reliance on single backward trajectory lines by generating 4D probabilistic origin density fields and testing candidate ranking persistence via Monte-Carlo environmental perturbations[1].
- **Suppression of Evidentiary Confirmation Bias:** Moving beyond one-sided proximity matching by evaluating contradictory alibi metrics alongside supporting indicators[1].
- **Mitigation of Lagrangian Backtracking Divergence:** Addressing the non-conservative nature of reverse advection through bidirectional forward-simulation consistency checks[1].
- **Structured AIS Integrity Auditing:** Replacing simplistic gap assumptions with kinematic reachability analysis and Unresolved Vessel Detection indexing for non-transponding hulls[1].
- **Suppression of Generative AI Hallucinations:** Restricting Large Language Models strictly to downstream, read-only summarization of immutable, cryptographically sealed Evidence JSON trees[1].

Open Challenges Requiring Ongoing Research and Engineering Attention

- **Constellation Revisit Latency vs. Weathering Half-Life:** Sentinel-1 C-band SAR operates with revisit intervals of 1 to 6 days depending on latitude and orbit track[15]. Because light operational discharges can disperse below radar detection thresholds within 12 to 24 hours under moderate sea states, sub-daily monitoring requires expanding data ingestion to emerging commercial small-satellite SAR constellations[2].
- **Sub-Mesoscale Coastal Hydrodynamics:** Global and regional Eulerian circulation models (such as CMEMS at 1/12 degree resolution) accurately resolve mesoscale structures but carry interpolation errors in shallow coastal boundary layers, tidal inlets, and complex archipelagoes where localized bathymetric shears dominate[2].
- **Volumetric Thickness Inversion from Monochromatic SAR:** Single-frequency C-band SAR measures capillary wave damping rather than absolute oil layer thickness[2]. Precise volumetric inversion across the full spectrum from thin sheens to heavy emulsions remains an open remote sensing challenge requiring multi-frequency SAR or hyperspectral cross-validation[2].
- **Advanced Non-Cooperative AIS Obfuscation:** While basic transponder shutoffs are audited via kinematic dead-reckoning, sophisticated electronic deception (such as coordinated GPS spoofing or dynamic MMSI swapping across multiple hulls) requires integration with satellite RF emitter geolocations and spaceborne ELINT feeds[1].
- **Investigative Ground-Truth Verification:** Under international maritime law (MARPOL 73/78 Annex I) and national frameworks, computational attribution provides actionable investigative intelligence for port-state inspections and interdiction, but full judicial prosecution often requires physical chemical fingerprinting matching bunker tanks[2].

Multi-Pod System Architecture and Modular Ownership

The Arpinia Odyssey platform enforces strict separation of concerns across three specialized operational tiers:

- **Pod A — SAR Intelligence:** Ingests Sentinel-1 Level-1 GRD SAR products; performs radiometric calibration, terrain correction, and speckle filtering; executes semantic segmentation benchmarking; runs contextual look-alike risk filtering; quantifies boundary uncertainty; cross-validates across diverse maritime scenes; and exports a schema-validated GeoJSON handoff payload with cryptographic integrity verification[1].
- **Pod B — Physics, AIS, Attribution & Evidence:** Ingests the validated GeoJSON payload from Pod A alongside ECMWF ERA5 atmospheric wind and CMEMS/INCOIS ocean hydrodynamic fields; executes stochastic Lagrangian drift simulations; reconstructs 4D probabilistic origin density fields; extracts and audits historical AIS trajectories; computes explainable multi-factor attribution scores; executes Monte-Carlo attribution stability analysis; validates candidate vessels via bidirectional forward simulations; manages attribution abstention logic; and compiles the final Evidence JSON tree with

tamper-evident cryptographic hashes[1].

- **Pod C — Presentation & Decision GIS:** Consumes the sealed Evidence JSON tree and geospatial layers via secure APIs; renders interactive web-based GIS dashboards, trajectory replays, and candidate vessel comparison tables; formats downloadable case dossiers and POLREP-aligned bulletins; and provides a responsive case management workspace for investigators. **Pod C functions strictly as an interactive visual interface and never computes attribution truth or modifies evidence states**[1].

Pod A: The Six Trustworthy SAR Intelligence Objectives

Pod A implements six engineering objectives designed to ensure that the spaceborne remote sensing layer delivers scientifically robust, uncertainty-bounded evidence to the downstream physics engine[1].

Objective A1: Robust SAR Oil-Slick Segmentation

Oil films dampen capillary and short gravity waves on the sea surface, reducing microwave radar backscatter and creating low-intensity dark formations on SAR imagery[9]. Rather than relying on simple intensity thresholding, Pod A trains and benchmarks at least two deep semantic segmentation architectures: a Residual U-Net (ResNet-34 backbone) and DeepLabV3+ with Atrous Spatial Pyramid Pooling (ASPP)[9].

The architectures are evaluated on Sentinel-1 C-band SAR datasets using standard segmentation metrics: Mean Intersection-over-Union (mIoU), Boundary Dice Coefficient, Precision, Recall, and F1-score[9]. The target output is a high-resolution slick polygon that accurately preserves complex morphology, including elongated discharge tails and feathering boundaries[1].

Objective A2: Look-Alike Suppression Engine

Radar backscatter reduction is non-unique to petroleum hydrocarbons[2]. Natural biogenic surfactant films, localized low-wind areas, atmospheric rain cells, internal waves, upwelling zones, and aerodynamic land/platform shadows produce low-backscatter dark patches that mimic oil slicks[2].

Pod A integrates a physical, contextual look-alike risk engine:

- **Contextual Wind Speed Evaluation:** Radar backscatter contrast requires moderate wind conditions (nominally between 3 m/s and 10 to 12 m/s) to generate Bragg scattering on clean water while preserving slick contrast[9]. Localized 10-meter wind speeds ($U_{10m} < 3\ \mathrm{m/s}$) harvested from ECMWF ERA5 or numerical weather predictions indicate elevated look-alike risk[1]. However, low wind alone does not automatically eliminate a slick, as genuine oil discharges can occur in calm seas; instead, low wind increases the contextual look-alike risk factor within a joint probabilistic confidence model[1].
- **Spatial Texture and Morphology:** Gray-Level Co-occurrence Matrix (GLCM) texture metrics (homogeneity, contrast, angular second moment) are evaluated alongside

dual-polarization (VV/VH) ratios[2]. Petroleum slicks exhibit distinct boundary contrast and structural coherence, whereas biogenic films display diffuse gradients and irregular spatial distributions[9].

- **Operational Metric Reporting:** The engine independently evaluates and reports true oil-slick segmentation accuracy alongside the Look-Alike False Positive Rate (FPR_{LA}), isolating its capability to separate non-petroleum dark formations[1].

Objective A3: Uncertainty-Aware Spill Boundary Estimation

Standard binary masks generated at a single threshold (e.g., $p \geq 0.50$) introduce artificial certainty regarding slick boundaries[1]. Because oil slicks exhibit varying thickness gradients near their periphery, radar damping decays gradually toward the background sea[2].
Pod A computes epistemic and aleatoric boundary uncertainty:

- **Epistemic Variance Modeling:** Evaluated via Monte-Carlo Dropout across model inferences, capturing model uncertainty along low-contrast edges[1].
- **Threshold Sensitivity Mapping:** Evaluates segmentation stability across a spectrum of decision thresholds ($\tau \in [0.30, 0.70]$), delineating a high-confidence core region ($p \geq 0.80$) and a boundary uncertainty envelope ($0.30 \leq p < 0.80$)[1].
- **Confidence Field Output:** Exports a spatial uncertainty gradient designed to inform downstream particle seeding density in Pod B[1].

Objective A4: Cross-Scene Generalization and Failure Analysis

Segmentation models often degrade significantly when evaluated on unseen geographic regions, altered radar incidence angles ($20^\circ \text{ to } 45^\circ$), or varying sea states[8].
Pod A incorporates cross-scene validation using independent Sentinel-1 SAR datasets, such as DARTIS (Eastern Mediterranean Sea) and CSIRO (Australian waters)[8]. Cross-dataset testing validates model transferability across open-ocean and complex coastal environments[8].
Pod A establishes an explicit Failure Mode Taxonomy, reporting operational performance limits under:

- **High-Wind Entrainment ($U_{10m} > 12\ \mathrm{m/s}$):** Severe wave action breaking up slicks and submerging oil droplets[7].
- **Steep Incidence Angles ($< 25^\circ$):** Elevated background backscatter noise reducing slick-to-sea contrast[12].
- **Nearshore Topographic Shadowing:** Coastal terrain occluding radar returns, mitigated through high-resolution coastline buffering[6].

Objective A5: Validated Geospatial Evidence Handoff to Drift Intelligence

Pod A packages its analytical outputs into a schema-validated GeoJSON handoff contract

designed directly for Pod B's hydrodynamic drift models[1].
The handoff payload contains:

- Case UUID and Sentinel-1 Scene Metadata (Orbit Number, Acquisition Timestamp UTC, Sensor Mode, Polarization)[1].
- Slick Geometry: Multi-polygon coordinates in WGS84 (EPSG:4326)[1].
- Morphological Attributes: Total Surface Area in square kilometers, Geometric Centroid, Major Axis Length, and Spatial Orientation[1].
- Observational Quality Proxies: Mean Radar Contrast ($\Delta\sigma_0$ dB), Look-Alike Risk Index, Epistemic Boundary Uncertainty Score, and Mean Segmentation Confidence[1].
- Data Provenance: Cryptographic SHA-256 integrity hash sealing the raw SAR scene, model weights version, and preprocessing configuration[1].

Objective A6: Evidence-Locked Local AI Interpretation

To eliminate the risk of generative artificial intelligence hallucinations in operational reporting, local natural language models are deployed strictly downstream as read-only translation interfaces[1].
An air-gapped Large Language Model (deployed locally via Ollama) ingests the finalized JSON evidence payload[1]. Under programmatic constraints:

- The LLM summarizes technical metrics (segmentation confidence, look-alike risk, boundary dispersion) into clear operational language[1].
- The model is strictly prohibited from calculating scores, modifying spatial coordinates, altering polygon boundaries, or assigning legal liability[1].
- If data quality is degraded, the LLM outputs an explicit caution recommending aerial reconnaissance or surface sampling[1].

Pod B: The Six Frozen Objectives for Physics, AIS, Attribution & Evidence

Pod B is designed as the scientific, physical, and forensic attribution engine of the platform, structured around six core capabilities[1].

Objective B1: Probabilistic Spill-Origin Reconstruction

Deterministic single-path backtracking is scientifically fragile, as numerical current and wind fields contain irreducible spatio-temporal errors[1]. Pod B's target architecture executes stochastic Lagrangian backward tracking using the OpenDrift framework[7]. In the target design, numerical particles will be seeded across the slick polygon, with spatial particle density weighted by Pod A's boundary uncertainty map[1].
The target model advects particles backward in time via coupled stochastic differential equations driven by ECMWF ERA5 atmospheric wind, CMEMS/INCOIS Eulerian currents, and wave drift parameterizations[7]. Turbulent diffusion is modeled via a random walk process, producing a 4D spatio-temporal probability density field $\mathcal{P}_{origin}(x, y, t)$ that delineates the probable release corridor rather than an unphysical point coordinate[1].

Objective B2: Spatio-Temporal AIS Correlation Engine

Pod B ingests historical terrestrial and satellite AIS data covering the domain and temporal window of the reconstructed origin plume[1]. It applies multi-stage spatial-temporal bounding filters to eliminate non-intersecting vessel traffic[1].

For surviving candidate vessels, the engine performs kinematic trajectory reconstruction, evaluating vessel speed, heading changes, and closest approach vectors relative to the evolving origin field $\mathcal{P}_{origin}$ [1]. Where AIS coverage gaps occur, kinematic dead-reckoning evaluates whether a vessel could have transited the origin envelope while unobserved[1].

Objective B3: Explainable Multi-Factor Vessel Attribution

To avoid opaque decision-making, candidate ranking is computed via an explainable multi-factor scoring function that evaluates distinct physical and behavioral indicators[1].

The candidate fitness score $\Phi(V_k)$ integrates:

- Spatial-Temporal Origin Overlap: Spatial coincidence with the high-probability core of $\mathcal{P}_{origin}$ [1].
- Geometric Alignment: Consistency between the vessel's historical heading vector and the slick's major axis orientation[1].
- Kinematic Feasibility: Operational vessel speed profiles and course stability during the estimated discharge interval[1].
- Weathering Compatibility: In advanced configurations, consistency between estimated transit timing and initial ADIOS oil weathering parameters (such as evaporative loss limits)[7].

Objective B4: Monte-Carlo Attribution Stability Engine

To determine whether candidate rankings are robust under environmental noise, Pod B is designed to execute a Monte-Carlo perturbation ensemble[1]. Rather than relying on an arbitrary fixed iteration count, the ensemble size M is selected experimentally using convergence testing (evaluating stability across $M \in \{50, 100, 250, 500\}$ until candidate rank distributions stabilize)[1].

Across perturbation runs, the engine injects controlled variations into wind velocity vectors ($\pm 15\%$), current directions ($\pm 10^{\circ}$), wind drift factors ($\gamma_{wind} \sim \mathcal{U}(0.02, 0.035)$), and release timestamps[1]. The Attribution Stability index $S(V_k)$ is defined as the empirical probability that candidate vessel V_k maintains the top ranking across the perturbation ensemble:

$$S(V_k) = \frac{1}{M} \sum_{m=1}^{M} \mathbb{I}\left(\arg\max_j \left[\Phi(V_j \mid \boldsymbol{\Theta}^{(m)})\right] = k\right)$$

Candidate prioritization requires satisfying a configurable stability threshold $\tau_{stability}$ (e.g., $S(V_k) \geq 0.75$, calibrated experimentally against historical test cases)[1].

Objective B5: Counter-Evidence & Bidirectional Validation

To prevent confirmation bias, Pod B systematically evaluates contradictory evidence alongside supporting factors[1].

- **Adversarial Alibi Scoring:** The system checks for contradictory indicators, including large Closest Point of Approach (CPA) distances, temporal transit outside the reconstructed release window, uncompromised AIS continuity with steady cruising speed, and conflicting forward dispersion patterns[1]. The net evidentiary balance is computed as:

$$E_{net}(V_k) = \sum_{i=1}^{P} w_i^+ \psi_i^+(V_k) - \sum_{j=1}^{Q} w_j^- \psi_j^-(V_k)$$

- **Bidirectional Forward Drift Validation:** For leading candidate vessels, the target architecture initiates a forward Lagrangian simulation from the vessel's candidate release state to the satellite acquisition time t_{SAR} [1]. The forward-simulated surface particle envelope $\mathcal{M}_{fwd}(V_k)$ is geometrically compared against the satellite-observed polygon $\mathcal{M}_{SAR}$ using the Intersection-over-Union (IoU) metric:

$$C_{bidi}(V_k) = \mathrm{IoU}\left(\mathcal{M}_{SAR}, \mathcal{M}_{fwd}(V_k)\right) = \frac{\mathrm{Area}\left(\mathcal{M}_{SAR} \cap \mathcal{M}_{fwd}(V_k)\right)}{\mathrm{Area}\left(\mathcal{M}_{SAR} \cup \mathcal{M}_{fwd}(V_k)\right)}$$

- **Attribution Abstention Protocol:** If Attribution Stability falls below operational thresholds or contradictory evidence dominates, Pod B triggers an explicit **INSUFFICIENT EVIDENCE** declaration, preventing premature conclusions under ambiguous environmental conditions[1].

Objective B6: Auditable Forensic Evidence Architecture

Pod B compiles analytical outputs, observational metadata, simulation trajectories, and scoring matrices into a unified, machine-readable Evidence JSON tree[1].

Every component is linked to a SHA-256 cryptographic provenance chain covering raw satellite

scene IDs, numerical model run versions, AIS feed checksums, and exact simulation parameters[1]. While cryptographic hashing provides tamper-evident integrity verification rather than independent legal proof, this frozen JSON artifact provides the single source of truth for downstream visual display, supporting reproducible institutional and investigative review[1].

Pod C: Presentation, Visualization, and Case Management UI/UX

Pod C provides the investigator-facing presentation layer, delivering a responsive, WebGIS-enabled workspace[1].

- **Interactive Spatial-Temporal WebGIS:** Renders satellite SAR polygons, boundary uncertainty buffers, metocean vector overlays (wind/currents), and dynamic 4D origin probability fields on an interactive Leaflet/Mapbox canvas[1].
- **Vessel Trajectory & Anomaly Playback:** Displays reconstructed historical AIS tracks, highlighted coverage gaps, dead-reckoning reachability corridors, and SAR-detected Unresolved Vessel positions with synchronized time-scrubbing controls[1].
- **Candidate Comparison & Evidence Inspector:** Presents ranked candidate vessel scorecards showing net evidentiary balance (E_{net}), Attribution Stability percentages (S_k), bidirectional overlap metrics (C_{bidi}), and side-by-side breakdowns of supporting versus contradictory alibi factors[1].
- **Case Dossier & Export Engine:** Formats and exports audit-ready PDF investigative dossiers and POLREP-aligned bulletins, embedding cryptographic SHA-256 hashes to verify tamper-evident data integrity[1].

Explicit Separation of Uncertainty Dimensions

To maintain scientific rigor and prevent misleading aggregations, the platform maintains separate, unmerged uncertainty metrics across every stage of the pipeline[1].

Uncertainty Dimension	Pipeline Origin	Mathematical / Physical Representation	Operational Interpretation
Segmentation Confidence	Pod A (SAR Vision)	Pixel-level Softmax Probability ($p \in$)[9]	Local model confidence in dark-pixel classification
Look-Alike Risk	Pod A (Context Engine)	Joint MetOcean & Textural Risk Score (R_{LA})[2]	Probability of non-oil origin (low wind, biofilm, rain)

Boundary Uncertainty	Pod A (Morphology)	Epistemic Dropout Variance & Threshold Sensitivity[1]	Spatial edge fuzziness and thickness transition width
MetOcean Forcing Quality	Pod B (Hydrodynamics)	Numerical Model Resolution & Interpolation Variance[2]	Confidence in current and wind forcing field accuracy
Origin Spatial Dispersion	Pod B (Lagrangian)	4D Probability Density Field $\mathcal{P}_{origin}(x, y, t)$ [cite: 1, 2]	Geographic area and time window of possible release
AIS Data Completeness	Pod B (Vessel Audit)	Transmission Continuity Index & Gap Duration (Q_{AIS})[1]	Reliability of transponder history vs. dead-reckoning
Attribution Score	Pod B (Evidence)	Multi-Factor Evidentiary Balance ($E_{net} \in$)[1]	Net balance of supporting vs. contradictory factors
Attribution Rank Stability	Pod B (Monte-Carlo)	Empirical Top-Rank Probability ($S_k \in$)[1]	Robustness of candidate ranking under noise injection
Bidirectional Consistency	Pod B (Forward Drift)	Geometric Overlap Metric ($C_{bidi} =$)[1]	Physical shape recovery from candidate release state

Mathematical Formulations and Hydrodynamic Governing Equations

The transport, dispersion, and weathering of marine hydrocarbon particles within the Lagrangian framework are governed by continuous-time stochastic differential equations coupled with empirical oceanographic parameterizations[7].

The displacement of each discrete Lagrangian oil particle p is formulated as:

$$dx_p(t) = \left[u_{current}\left(x_p, t\right) + \gamma_{wind}\, u_{10m}\left(x_p, t\right) + u_{Stokes}\left(x_p, t\right)\right] dt + u'_{turbulent}\, dt$$

where $u_{current}$ denotes the Eulerian surface ocean current velocity vector obtained from hydrodynamic circulation models, u_{10m} is the 10-meter atmospheric wind vector, γ_{wind} represents the empirical surface wind drift factor (nominally parameterized between 0.020 and 0.035), and u_{Stokes} is the wave-induced Stokes drift vector[7]. Sub-grid scale turbulent dispersion is modeled as a random walk stochastic process:

$$u'_{turbulent}\, dt = \sqrt{2K_h \Delta t}\, R_n$$

where K_h represents the horizontal eddy diffusivity coefficient and $R_n \sim \mathcal{N}(0, \mathbf{I})$ is a bivariate standard normal random vector[14].
In advanced extensions, the physical transformation of the slick is governed by coupled ADIOS weathering formulations, wherein the evaporative mass loss fraction F_{evap} over exposure time t under sea surface temperature T_s and wind speed U_{wind} is computed as:

$$F_{evap} = \frac{T_s}{K_e} \ln\left(1 + \frac{K_e K_a \theta}{T_s} \exp\left(A - \frac{B T_b}{T_s}\right)\right)$$

where K_e, K_a, A, and B represent distillation constants for specific petroleum fractions, T_b is the initial boiling point, and θ is the evaporative exposure parameter[7].

Operational Alignment with Indian Statutory Frameworks and Response Protocols

To provide practical utility for national maritime law enforcement, the platform is integrated into the statutory and operational architecture governing the Indian Exclusive Economic Zone[2].
The legal foundation for maritime pollution interdiction in India is established under Part XI-A of the Merchant Shipping Act 1958, which incorporates India's obligations under MARPOL 73/78 Annex I[3]. Sections 356G and 356K empower the Indian Coast Guard as the Central Coordinating Authority to inspect vessels, detain ships suspected of illegal operational discharges, and levy statutory penalties[3]. Under this framework, evidence presented to maritime

Evidence presented to maritime authorities should be scientifically reproducible and free from unverifiable heuristics. The platform is designed for alignment with the National Oil Spill Disaster Contingency Plan (NOS-DCP) hierarchy rather than implying an existing institutional integration. It can consume compatible ocean-state, sea-surface-temperature, current, and wave products from authoritative sources such as INCOIS, execute source reconstruction, and generate investigation-support outputs structured for maritime-response workflows. Analytical findings can be exported in a POLREP-aligned structure for downstream human review and operational adaptation.

POLREP Component	Section Identifier	Operational Content and Ingested System Evidence
POLWARN	Part I: Incident Identification	Automated Case ID, Observation Timestamp (UTC), Satellite Platform (Sentinel-1 SAR)[1]
(Pollution Warning)	Geographic Coordinates	Exact Latitude/Longitude coordinates of detected slick centroid and polygon vertices[1]
	Physical Slick Metrics	Surface area in km^2, estimated thickness proxy, computed volume envelope[1]
POLINFO	Part II: Metocean & Drift	INCOIS ocean current vectors, ERA5 wind velocity, sea state, water temperature[1]
(Pollution Information)	Probabilistic Origin Field	4D reconstructed source corridor coordinates and release timestamp window[1]

	Look-Alike Risk Assessment	Look-alike risk score (R_{LA}), local wind-speed verification, textural contrast[2]
POLFAC	Part III: Candidate Vessel Matrix	Ranked Candidate Vessels (IMO, MMSI, Vessel Name, Flag State, Call Sign)[1]
(Attribution & Action)	Forensic Attribution Metrics	Attribution Stability index (S_k), Bidirectional Score (C_{bidi}), AIS Continuity (Q_{AIS})[1]
	Evidentiary Balance	Explicit breakdown of Supporting Factors (ψ^+) vs. Contradictory Alibi Factors (ψ^-)[1]
	Tamper-Evident Integrity	SHA-256 hash sealing raw data metadata, model parameters, and trajectory states[1]

Technical Benchmarking Matrix and Phased Implementation Roadmap

Evaluating the performance of the forensic framework requires a standardized benchmarking protocol that evaluates each subsystem across development milestones[1].

Subsystem Module	Target Metric	Operational Threshold	Evaluation Protocol and Dataset
SAR Segmentation (Pod A)	Mean Intersection-over-Union (mIoU)	$\geq$ (Standard), $\geq$	Sentinel-1 SAR imagery (DARTIS & CSIRO datasets)[8]

		(Cross-scene)	
	Boundary Dice Coefficient	$\geq$ (Standard), $\geq$ (Cross-scene)	Evaluated against expert marine annotations[8]
Look-Alike Engine (Pod A)	Look-Alike False Positive Rate (FPR_{LA})	$\leq$ across low-wind / film scenes	Curated benchmark of non-oil dark formations[2]
Boundary Uncertainty (Pod A)	Epistemic Boundary Variance	Continuous spatial gradient field	Evaluated via MC Dropout and threshold sweeps[1]
Hydrodynamic Engine (Pod B)	Lagrangian Trajectory Error	$\leq$ per 24 hours of drift	Calibrated against surface drifting buoy datasets[7]
Attribution Engine (Pod B)	Primary Rank Stability (S_k)	$\geq$ under experimental convergence	Simulated discharge scenarios with noise injection[1]
	Adversarial Alibi Detection Rate	Precision and recall reported on controlled AIS fixtures; thresholds set from validation results	Dense synthetic AIS traffic corridors[1]
Provenance Layer (Pod B)	Cryptographic Integrity Verification	100% of unchanged validation artifacts reproduce recorded SHA-256; altered fixtures fail verification	End-to-end evidence chain verification on validation suites[1]

Phased Implementation Roadmap

To maintain engineering discipline and prevent architectural overengineering, development is structured into four sequential phases[1]:

- **Phase 1 — Physical Baseline MVP:**

- Ingest Sentinel-1 Level-1 GRD SAR imagery and apply standard preprocessing[1].
- Train and benchmark baseline U-Net and DeepLabV3+ segmentation models on reference scenes[9].
- Export validated slick polygon geometries to OpenDrift[1].
- Execute baseline deterministic Lagrangian hindcasting using ECMWF ERA5 wind and CMEMS ocean current fields[1].
- Perform basic spatial-temporal candidate vessel intersection against historical AIS tracks[1].

- **Phase 2 — Probabilistic Attribution:**
  - Integrate Pod A's boundary uncertainty and contextual look-alike risk modeling into the handoff payload[1].
  - Deploy stochastic particle tracking to construct 4D Probabilistic Origin Fields $\mathcal{P}_{origin}(x, y, t)$ [1].
  - Implement Pod B's explainable multi-factor attribution scoring function[1].
- **Phase 3 — Robustness & Counter-Evidence:**
  - Deploy Pod B's Monte-Carlo Attribution Stability Engine with experimental convergence testing ($M \in \{50, 100, 250, 500\}$)[1].
  - Implement adversarial counter-evidence and alibi scoring alongside supporting metrics[1].
  - Incorporate AIS integrity checks, dead-reckoning audits, and SAR Unresolved Vessel Detection indexing[1].
  - Execute bidirectional forward-simulation consistency validation (C_{bidi})[1].
  - Implement the Attribution Abstention Protocol for low-stability or high-contradiction cases[1].
- **Phase 4 — Evidence Serialization & Presentation:**
  - Serialize all findings into the Evidence JSON schema sealed with SHA-256 cryptographic provenance hashes for tamper-evident verification[1].
  - Expose backend APIs to Pod C's interactive WebGIS case management dashboard and PDF generation engine[1].
  - Integrate downstream, air-gapped local LLM (via Ollama) strictly for read-only POLREP-aligned bulletin generation[1].

Strategic Synthesis and System Conclusions

The primary contribution of the Arpinia Odyssey platform is bridging the divide between automated satellite detection and scientifically defensible investigative attribution[1]. Existing operational systems provide critical monitoring and forecasting capabilities, but attribution under environmental uncertainty remains an active challenge across the maritime domain[1].

By enforcing strict pod ownership, the architecture establishes an auditable intelligence pipeline[1]. Pod A delivers SAR segmentation, contextual look-alike risk suppression, and boundary uncertainty estimation[1]. Pod B is designed to transform this observational input into a

probabilistic origin corridor, audit vessel trajectories, test ranking stability across Monte-Carlo environmental perturbations, challenge findings with adversarial alibis, and validate candidate vessel releases bidirectionally[1]. Pod C provides an investigator-friendly GIS interface without computing attribution truth, while downstream generative language models remain strictly locked to finalized evidence[1].

Aligned with the Indian statutory mandate under Part XI-A of the Merchant Shipping Act 1958 and the NOS-DCP, the platform equips maritime authorities with an objective, reproducible decision-support system to enforce environmental compliance across national and international waters[1].

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APPENDIX A - REFINEMENT & VALIDATION POLICY

1. Claim discipline

Competitor limitations are stated only as capabilities not identified in the public documentation reviewed. Candidate-vessel language is used throughout; the system supports investigation and does not declare guilt.

2. Benchmark policy

Performance thresholds are not frozen before experiments. Segmentation, trajectory, attribution, and stability metrics must be reported from controlled validation sets, cross-scene tests, or convergence studies.

3. Evidence-integrity policy

SHA-256 is used for tamper-evident integrity verification. Acceptance testing checks that unchanged artifacts reproduce their recorded hashes and intentionally altered fixtures fail verification. Hashing is not treated as independent legal proof.

4. Institutional-alignment policy

NOS-DCP, INCOIS, Indian Coast Guard, MRCC, and POLREP references describe design alignment and compatible output structures unless a formal operational integration has been independently established.

5. Source-hardening policy

Internal notes, mirrors, aggregators, and discovery sources may support reconnaissance, but final scientific and competitive claims should preferentially cite original peer-reviewed publications and official institutional documentation.

Primary-source anchors for final literature hardening

- SkyTruth Cerulean Methods - https://skytruth.org/cerulean/methods

- EMSA CleanSeaNet - https://emsa.europa.eu/

- INCOIS Online Oil Spill Advisory (OOSA) - https://oosa.incois.gov.in/

- Global Fishing Watch API / vessel intelligence - https://globalfishingwatch.org/our-apis/documentation/

- OpenDrift v1.0, Geosci. Model Dev. 11, 1405-1420 (2018) - DOI: 10.5194/gmd-11-1405-2018

- ECMWF ERA5 official dataset documentation - https://www.ecmwf.int/

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